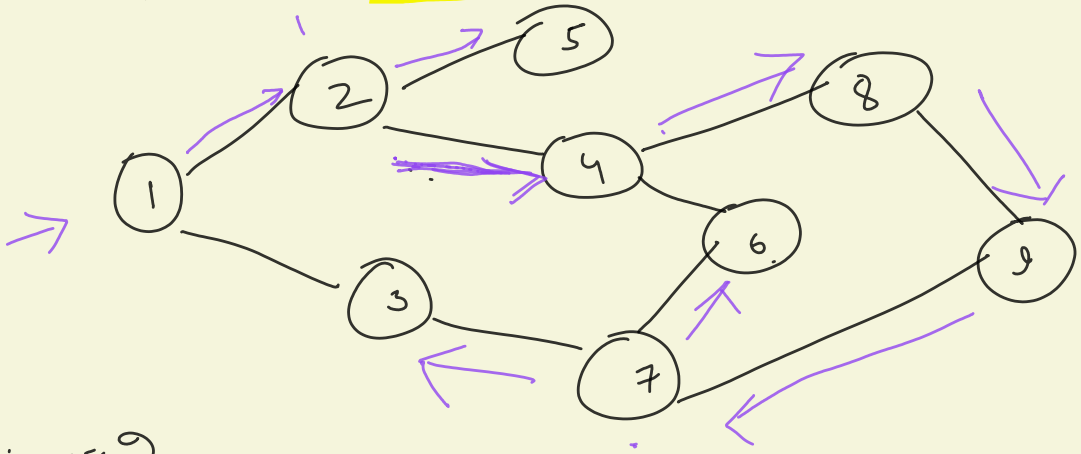
DFS :- Depth First Search



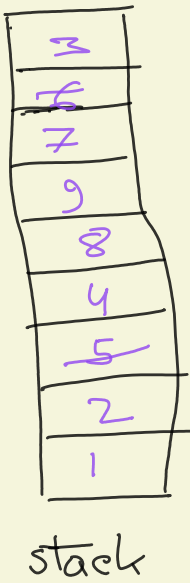


Size=9

visited [ ] * vertices

T	T	T	T	T	T	T	T	T
1	2	3	4	5	6	7	8	9

1 → 2 → 5 → 4 → 8 → 9 → 7 → 6 → 3

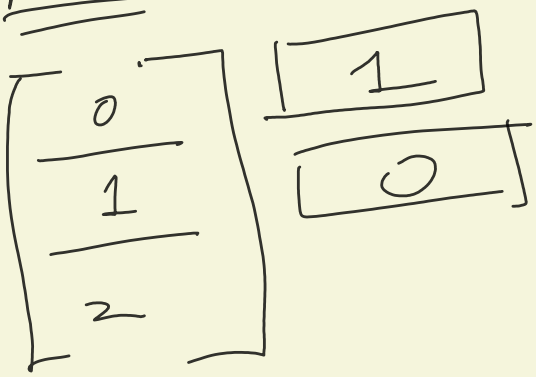


stack

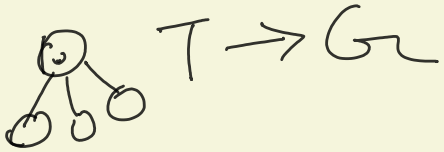
matrix

	0	1	2
0	0	1	0
1	1	0	0
2	0	2	2

AL



Note :- ① Every tree is a graph, but not every graph is tree.



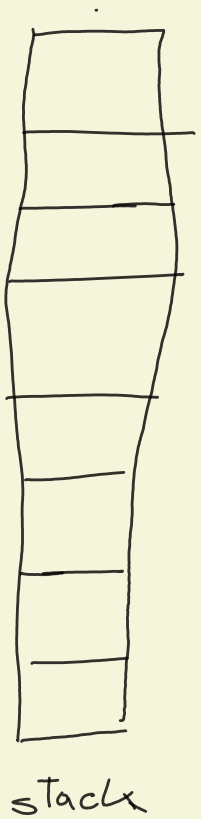
② o/p can varry person to person.

(3) starting point can vary also.

Note :- If any connected graph doesn't forms a circle, then it's a tree

logic

visited = | F | F | F | F | F | F | F | F |
0 1 2 3 4 5 6 7




```
item = stack.pop
if visited[item] == False
    print()
    visited = True
    for i in mat[node]
        if visited[i] == False
            stack.push(i)
```

Note:-

* if not visited node
  - print
  - visited True
  - stack rest of nabours